Robotics

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This is a set of notes covering interesting concepts I find while studying the Modern Robotics textbook. I will try to bring analytic theory whenever applicable.

Textbooks Used to Develop these Notes:

1. ***Modern Robotics*** - *Kevin Lynch, Frank Park*
2. ***Calculus on Manifolds*** - *Michael Spivak*
3. ***Mathematical Analysis*** - *Tom Apostol*

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Holonomic Constraints

1.1 Basic Multivariable Analysis

Note: The following notes may not be entirely rigorous! I have attempted to write them logically to the best of my abilities so that they follow a structured narrative.

We begin with the exploration of multivariate analysis. When we deal with a multivariate function in $\mathbb{R}$, we deal with some map of the form $f : \mathbb{R}^n \to \mathbb{R}^m$. We can define this more formally via vector operations as the following:

$$f\left(\begin{bmatrix} x_1 \\ . \\ . \\ . \\ x_n \end{bmatrix}\right) = \begin{bmatrix} f_1(x_1, x_2, ..., x_n) \\ . \\ . \\ . \\ f_m(x_1, x_2, ..., x_n) \end{bmatrix} \tag{1}$$

such that $\forall j \in [1, m] : f_j : \mathbb{R}^n \to \mathbb{R}$. Our original definition of the derivative in $\mathbb{R}$ was:

$$f'(x) \coloneqq \lim_{z \longrightarrow x} \frac{f(z) - f(x)}{z - x}$$

or equivalently

$$\lim_{z \longrightarrow x} \frac{f(z) - f(x) - f'(x)(z - x)}{z - x} = 0$$

Using the substitution $h = z - x$, we also get the equivalent definition:

$$\lim_{h \longrightarrow 0} \frac{f(x + h) - f(x) - f'(x)h}{h} = 0$$

In which case, our $\epsilon - \delta$ definition would be:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{|f(x + h) - f(x) - f'(x)h|}{|h|} < \epsilon$$

However if we try to plainly substitute the definition for a function $f : \mathbb{R}^n \to \mathbb{R}^m$ into the first definition, we get "division of two vectors" which is not a defined operation for vector spaces (they are only abelian groups under addition).

We consider the following lemma in order to understand how to extend the derivative to the multivariate case:

Lemma 1.1. *$f'(x)$ exists* $\iff \left\{\frac{1}{\delta} \sup_{z \in B(x,\delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)|\right\}_{\delta > 0} \to 0$

Proof. $\implies$: If $f'(x)$ exists, then:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left|\frac{f(z) - f(x)}{z - x} - f'(x)\right| < \epsilon$$

$$\implies \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leqslant \delta_0, \forall z \in B(x, \delta) \setminus \{x\} : \left|\frac{f(z) - f(x)}{z - x} - f'(x)\right| < \epsilon$$

$$\implies |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot |z - x| < \epsilon \cdot \delta$$

Because this is true for all z in the punctured δ-ball around x, we have an upper bound, so we take the supremum:

$$\implies \sup_{z\in B(x,\delta)\setminus\{x\}} |f(z)-f(x)-f'(x)(z-x)| < \epsilon\cdot\delta$$

And so we rearrange the terms to get a δ-net that converges such that:

$$\forall\epsilon>0, \exists\delta_0>0, \forall\delta\leqslant\delta_0 : \frac{1}{\delta}\sup_{z\in B(x,\delta)\setminus\{x\}} |f(z)-f(x)-f'(x)(z-x)| < \epsilon$$

Thus proving the forward direction.

$\impliedby$: Now for the converse, consider:

$$\forall\epsilon>0, \exists\delta_0>0, \forall\delta\leqslant\delta_0 : \sup_{z\in B(x,\delta)\setminus\{x\}} |f(z)-f(x)-f'(x)(z-x)| < \epsilon\cdot\delta$$

Now for any arbitrary δ-ball around x where $\delta\leqslant\delta_0$, we have that:

$$\sup_{z\in B(x,\delta)\setminus\{x\}} |f(z)-f(x)-f'(x)(z-x)| < \epsilon\cdot\delta$$

$$\implies \forall z\in B(x,\delta)\setminus\{x\} : \frac{1}{\delta}|f(z)-f(x)-f'(x)(z-x)| < \epsilon$$

$$\implies \forall z\in B(x,\delta_0)\setminus B(x,\delta) : \frac{|f(z)-f(x)-f'(x)(z-x)|}{|z-x|} < \frac{1}{\delta}|f(z)-f(x)-f'(x)(z-x)| < \epsilon$$

Since δ was arbitrary as long as it was less than or equal to δ_0, we observe:

$$\forall\epsilon>0, \exists\delta_0>0, \forall\delta\leqslant\delta_0, \forall z\in B(x,\delta_0)\setminus B(x,\delta) : \left|\frac{f(z)-f(x)}{z-x}-f'(x)\right| < \epsilon$$

$$\iff \forall\epsilon>0, \exists\delta_0>0, \forall z\in \bigcup_{\delta\in(0,\delta_0]} (B(x,\delta_0)\setminus B(x,\delta)) : \left|\frac{f(z)-f(x)}{z-x}-f'(x)\right| < \epsilon$$

$$\iff \forall\epsilon>0, \exists\delta_0>0, \forall z\in B(x,\delta_0)\setminus \bigcap_{\delta\in(0,\delta_0]} B(x,\delta) : \left|\frac{f(z)-f(x)}{z-x}-f'(x)\right| < \epsilon$$

Now because $\mathbb{R}$ is a T_1 space, we see that the intersection in the previous statement is equivalent to $\{x\}$, so we get:

$$\forall\epsilon>0, \exists\delta_0>0, \forall z\in B(x,\delta_0)\setminus\{x\} : \left|\frac{f(z)-f(x)}{z-x}-f'(x)\right| < \epsilon$$

A useful equivalent statement is:

$$\forall\epsilon>0, \exists\delta_0<\epsilon, \forall\delta\leqslant\delta_0, \forall z\in B(x,\delta)\setminus\{x\} : \frac{1}{\delta}|f(z)-f(x)-f'(x)(z-x)| < \epsilon$$

□

Now essentially this supremum δ-net, similar to oscillation is the worst case net. The important part of this result is the idea of linear approximation, such that:

$$f'(x)(z-x)+f(x)-\epsilon^2 < f(z) < f'(x)(z-x)+f(x)+\epsilon^2$$

or as we write it in calculus $f(z)\approx f(x)+f'(x)(z-x)$, which is the tangent line approximation.

We can apply this approximation idea to the multivariate case, but first we introduce the following notion:

Definition 1.1. *Given some* $f : \mathbb{R}^n \to \mathbb{R}^m$ *in the form of equation* (1)*, then for some* $i \in [1, m], j \in [1, n]$*:*

$$\frac{\partial f_i(x)}{\partial e_j} := \lim_{h \to 0} \frac{f_i(x + e_j h) - f_i(x)}{h}$$

where $\{e_1, \cdots, e_n\}$ *is the canonical basis of* $\mathbb{R}^n$*. This is the definition of the partial derivative.*

Since each $f_i : \mathbb{R}^n \to \mathbb{R}$, we see that the partial derivative is scalar valued. We now show that it admits the linear approximation property. We can rewrite our limit definition as:

$$\lim_{h \to 0} \left(\frac{f_i(x + e_j h) - f_i(x)}{h} - \frac{\partial f_i(x)}{\partial e_j} \right) = \lim_{h \to 0} \frac{f_i(x + e_j h) - f_i(x) - (\frac{\partial f_j(x)}{\partial e_j})h}{h} = 0$$

So with an argument similar to Lemma 1.1, we see that $f_i(z) \approx f_i(x) + \left(\frac{\partial f_i}{\partial e_j}\right) h$. Now observe that:

$$f(z) := \begin{bmatrix} f_1(z) \\ . \\ . \\ . \\ f_m(z) \end{bmatrix}$$

Lemma 1.2. *For all* $i \in [1, m]$ *and some* $x \in \mathbb{R}^n$ *and an* e_j*:*

$$\frac{\partial f_i(x)}{\partial e_j} \text{ exist}$$

if and only if:

$$\frac{\partial f(x)}{\partial e_j} \text{ exists}$$

Proof. $\Longrightarrow$: We claim that the partial derivative of f takes the following form:

$$\frac{\partial f(x)}{\partial e_j} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_j} \\ . \\ . \\ . \\ \frac{\partial f_m(x)}{\partial e_j} \end{bmatrix}$$

We have some δ_i given by the existence of each partial derivative, so we define $\delta_0 := \min\{\delta_i : 1 \leqslant i \leqslant n\}$. Then for all $h \in B(0, \delta_0)$:

$$\frac{||f(x + e_j h) - f(x) - (\frac{\partial f(x)}{\partial e_j})h||}{|h|} \leqslant \frac{1}{|h|} \sum_{i=1}^{m} \left| f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j}\right) h \right| < m\epsilon$$

where we used the inequality that the $||v|| \leqslant \sum |v_i|$.

$\Longleftarrow$: We claim that:

$$\frac{\partial f_i(x)}{\partial e_j} = e_i \cdot \frac{\partial f(x)}{\partial e_j}$$

We now use the inequality $|v_i| \leqslant ||v||$ and find that $\forall i \in [1, m]$:

$$\frac{|f_i(x + e_j h) - f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j})h|}{|h|} = \frac{|e_i \cdot f(x + e_j h) - e_i \cdot f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j})h|}{|h|}$$

$$= \frac{|e_i \cdot (f(x + e_j h) - f(x) - \frac{\partial f(x)}{\partial e_j} h)|}{|h|} \leqslant \frac{||f(x + e_j h) - f(x) - (\frac{\partial f(x)}{\partial e_j})h||}{|h|} < \epsilon$$

so the same δ suffices.

Now once again with an argument similar to Lemma 1.1, we can extend the idea of linear approximation and say that for all z such that $z = x + e_j h$:

$$f(z) \approx f(x) + \left(\frac{\partial f(x)}{\partial e_j} \right) h$$

□

Notice that the proofs did not use any properties of the basis vector e_j itself, so we can extend the definition:

Definition 1.2. *Given some $f : \mathbb{R}^n \to \mathbb{R}^m$ in the form of equation (1), then for some $v \in \mathbb{R}^n$ and $i \in [1, m], j \in [1, n]$:*

$$\frac{\partial f_i(x)}{\partial v} := \lim_{h \to 0} \frac{f_i(x + hv) - f_i(x)}{h}$$

and similarly for f. Such a derivative is called the Gateaux derivative.

We can extend the Gateaux derivative to obtain the Frechet derivative. The Gateaux derivative limited itself to a specific direction of the vector v (and then used h as the variable for the limit).

Definition 1.3. *The Frechet derivative of a function f at $x \in \mathbb{R}^n$ is:*

$$Df(x) : \lim_{v \to 0} \frac{||f(x + v) - f(x) - (Df(x))(v)||}{||v||} = 0$$

or equivalently:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0, \delta) : \frac{||f(x + v) - f(x) - (Df(x))(v)||}{||v||} < \epsilon$$

where the norm induces a metric so that $v \in B(0, \delta)$ (in this case 0 is the 0 vector), is the same as saying $||v|| \in B(0, \delta)$ or equivalently $||v|| < \delta$.

The form of the limit is similar to the idea of linear approximation. Since $v \in \mathbb{R}^n$, then we must have that $Df(x) \in \mathcal{M}_{m \times n}(\mathbb{R})$. If the function is Frechet differentiable, then $Df(x)$ is the Jacobian:

$$Df(x) = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_1} & \cdots & \frac{\partial f_1(x)}{\partial e_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x)}{\partial e_1} & \cdots & \frac{\partial f_m(x)}{\partial e_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f(x)}{\partial e_1} & \cdots & \frac{\partial f(x)}{\partial e_n} \end{bmatrix}$$

Proof. If we chose the family of vectors $\{he_j : h \in \mathbb{R} \wedge j \in [1, n]\}$, then differentiability tells us:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{||f(x + he_j) - f(x) - h(Df(x))e_j||}{|h|} < \epsilon$$

telling us that:

$$\frac{\partial f(x)}{\partial e_j} = Df(x)e_j$$

which is the j-th column of the Jacobian matrix. □

Notice that Lemma 1.2 was a statement on equivalence; however, the converse is not necessarily true for the Jacobian matrix. The total differentiability of f at x gives us the existence of partial derivatives, but not necessarily the converse. In order to prove this, we require the following concept:

1.1.1 MVT in the Multivariable Case

Theorem 1.1. *Given a function $f : \mathbb{R} \to \mathbb{R}$ that is continuous on the closed interval $[a, b]$ and differentiable on (a, b), then:*

$$\exists z \in (a, b) : f'(z) = \frac{f(b) - f(a)}{b - a}$$

Proof. Since f is continuous on a compact interval, it achieves the extrema by EVT. Now consider the function:

$$h(x) = \frac{f(b) - f(a)}{b - a}(x - a) - f(x) + f(a)$$

which at both $x = a$ and $x = b$, we have that $h(x) = 0$, so by Rolle's Theorem, there exists some extrema within the interval, so that:

$$\exists z \in (a, b) : h'(z) = 0 \implies f'(z) = \frac{f(b) - f(a)}{b - a}$$

□

If we try to apply this proof to a function $f : \mathbb{R}^n \to \mathbb{R}$, then in order to apply Theorem 1.1, we require some compact interval given f is continuous.

Now, the following lemma will be useful in exploring MVT:

Lemma 1.3. *Any open, convex subset $U \subseteq \mathbb{R}^n$ is path connected.*

Proof. Openness does not play an important role here. We can simply take the continuous function:

$$f(\alpha) = \alpha x + (1 - \alpha)y$$

for some fixed $x, y \in U$. By convexity, we know $\mathrm{Ran}(f) \subseteq U$. □

Lemma 1.4. *An open ball $B(x, r)$ is convex.*

Proof. Take $y, z \in B(x, r)$, then:

$$|\alpha y + (1 - \alpha)z - x| = |\alpha(y - x) + (1 - \alpha)(z - x)| \leqslant \alpha r + (1 - \alpha)r = r$$

□

Theorem 1.2. *Given $f_i : \mathbb{R}^n \to \mathbb{R}$, we have that for some open, convex subset $U \subseteq \mathbb{R}^n$ (on which f_i is differentiable) and given two points a, b:*

$$\exists z \in U : Df_i(z)(b - a) = f_i(b) - f_i(a)$$

where $Df_i(z)$ is the gradient vector at z.

Proof. Since U is convex, we can take the function for $t \in [0,1]$:

$$h(t) = tb + (1-t)a$$

such that $h(0) = a$ and $h(1) = b$. Then applying Theorem 1.1 to the function $f_i \circ h$ (which too is continuous on $[0,1]$ and differentiable on $(0,1)$), we see that:

$$\exists c \in (0,1) : (f_i \circ h)'(c) = \frac{f_i(h(1)) - f_i(h(0))}{1-0} = f_i(b) - f_i(a)$$

and $(f_i \circ h)'(c) = Df_i(c) \cdot h'(c) = Df_i(c)(b-a)$, where:

$$Df_i(x) = \begin{bmatrix} \frac{\partial f_i(x)}{\partial e_1} & \cdots & \frac{\partial f_i(x)}{\partial e_n} \end{bmatrix}$$

We used the multivariate chain rule in this proof, where $(f \circ h)'(t) = Df(h(t))Dh(t)$. In this case, we defined:

$$h(t) = \begin{bmatrix} tb_1 + (1-t)a_1 & \cdots & tb_n + (1-t)a_n \end{bmatrix}^T$$

$\square$

Corollary 1.1. *If $\exists k \in \mathbb{R}, \exists j \in [1,n] : b - a = ke_j$ and the partial derivative of f_i exists in the e_j direction, then:*

$$\exists z \in span(e_j) : \frac{\partial f_i(z)}{\partial e_j}(b-a) = f_i(b) - f_i(a)$$

Proof. Notice this corollarly does not require full differentiability of f_i (which means it does not require the gradient to be the total derivative of f at said point). $\square$

Lemma 1.5. *The function $f : \mathbb{R}^n \to \mathbb{R}^m$ is totally differentiable at a point if:*

1. *The partial derivatives exist*
2. *The partial derivatives are continuous at that point*

Proof. We want to show:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0,\delta) : \frac{||f(x+v) - f(x) - (Df(x))v||}{||v||} < \epsilon$$

The expression $(Df(x))(v)$ gives us the following vector:

$$(Df(x))(v) = \begin{bmatrix} \sum_{i=1}^n \frac{\partial f_1(x)}{\partial e_i} v_i \\ \vdots \\ \sum_{i=1}^n \frac{\partial f_m(x)}{\partial e_i} v_i \end{bmatrix}$$

and we see that:

$$\frac{||f(x+v) - f(x) - (Df(x))(v)||}{||v||} \leqslant \frac{1}{||v||} \sum_{j=1}^m \left| f_j(x+v) - f_j(x) - \sum_{i=1}^n \frac{\partial f_j(x)}{\partial e_i} v_i \right|$$

Continuity of the partial derivatives give us:

$$\forall \epsilon > 0, \exists \delta_{ji} > 0, \forall v \in B(0,\delta_{ji}) : \left| \frac{\partial f_j(x+v)}{\partial e_i} - \frac{\partial f_j(x)}{\partial e_i} \right| < \epsilon$$

We set $\delta := \min\{\delta_{ji} : 1 \leqslant j \leqslant m \wedge 1 \leqslant i \leqslant n\}$. Then given some $v \in B(0,\delta)$, by Lemma 1.4, we have that the open ball is convex, so we can apply MVT to each pair $f_j(x+v) - f_j(x)$.

Since $\square$

1.1.2 Multivariable Chain Rule

1.2 Inverse and Implicit Function Theorems

1.3 Configuration Spaces

The idea of configuration spaces is intertwined with the idea of manifolds, which I have not studied, so for this section we will limit ourselves to Euclidean configuration spaces.